

Review

Trends and Challenges in Real-Time Stress Detection and Modulation: The Role of the IoT and Artificial Intelligence

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Abstract

The integration of Internet of Things (IoT) devices and Artificial Intelligence (AI) has opened new frontiers in mental health, particularly in stress detection and management. This review explores the current literature, examining how IoT-enabled wearables, sensors, and mobile applications, combined with AI algorithms, are utilized to monitor physiological and behavioral indicators of stress. Advancements in real-time stress detection, personalized interventions, and predictive modeling are highlighted, alongside a critical evaluation of existing technologies. While significant progress has been made in the field, several limitations persist, including challenges with the accuracy of stress detection, the scalability of solutions, and the generalizability of AI models across diverse populations. Key challenges are further analyzed, such as ensuring data privacy and security, achieving seamless technological integration, and advancing model personalization to account for individual variability in stress responses. Addressing these challenges is essential to developing robust, ethical, and user-centric solutions that can transform stress management in mental healthcare. This review concludes with recommendations for future research directions aimed at overcoming current barriers and enhancing the effectiveness of IoT- and AI-driven approaches to stress management.

Keywords: IoT; Artificial Intelligence; stress detection; wearable sensors; mental health; real-time monitoring; machine learning; affective computing



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1. Introduction

1.1. Stress Pervasiveness and the Need for Objective Monitoring

Stress is a ubiquitous phenomenon in modern society, exerting significant influence on both individual well-being and public health [1]. Chronic or poorly managed stress is involved in a wide range of adverse health outcomes, including cardiovascular diseases, metabolic disorders, weakened immune function, and various psychological or behavioral conditions such as depression and anxiety [1]. The cumulative impact of daily stressors underscores the critical need for effective stress management strategies [2]. However, traditional methods for assessing stress, primarily relying on self-report questionnaires and interviews, suffer from inherent limitations. These methods are often subjective, prone to recall bias, influenced by the individual's current mood or self-awareness, and typically provide only retrospective or snapshot views rather than continuous, real-time insights into stress dynamics [3]. This lack of objective, granular, and timely data hinders the development and delivery of personalized and effective stress management interventions. The advent of digital technologies, particularly the convergence of the Internet of Things

(IoT) and Artificial Intelligence (AI), presents a transformative opportunity to overcome these limitations by enabling the continuous, objective, and real-time monitoring of physiological and behavioral indicators associated with stress [3].

1.2. The Convergence of the IoT and AI in Mental Health Technology

The synergy between the IoT and AI is rapidly reshaping the landscape of mental health technology. The IoT encompasses a network of interconnected physical objects—devices, vehicles, buildings, and other items embedded with electronics, software, sensors, actuators, and network connectivity—that enables these objects to collect and exchange data [4]. In the context of stress monitoring, the IoT facilitates ubiquitous data acquisition through a variety of sensors, including wearable devices (e.g., smartwatches and chest straps), mobile phones, and potentially environmental sensors integrated into smart homes or workplaces [5]. These sensors capture continuous streams of physiological data (e.g., heart rate, skin conductance, and temperature) and behavioral data (e.g., activity levels, sleep patterns, and location) [2].

While the IoT provides the means for data collection, AI, encompassing machine learning (ML) and deep learning (DL), furnishes the analytical power required to interpret the complex, high-dimensional, and often noisy data streams generated by these devices [3]. ML/DL algorithms can identify subtle patterns, correlations, and anomalies within physiological and behavioral data that correspond to different stress levels or affective states. This capability moves beyond simple thresholding to enable nuanced classification, prediction, and the personalization of stress assessment. The convergence of the IoT and AI thus paves the way for a paradigm shift in mental healthcare, moving from reactive treatment towards proactive, personalized, and continuous stress management strategies delivered just in time, potentially improving health behaviors and overall well-being [2].

1.3. Objectives and Structure of the Review

The primary objective of this scientific review is to critically synthesize the current body of literature concerning the application of the IoT and AI for the real-time detection and modulation of psychological stress. This review focuses on identifying key trends, evaluating the employed technologies (sensors and algorithms), analyzing the persistent challenges, and outlining promising future research directions. The analysis is grounded in the provided bibliography, encompassing research articles, reviews, and dataset descriptions pertinent to the field. It is important to acknowledge that a number of the initially provided literature sources were inaccessible at the time of the review (e.g., [6]); therefore, the synthesis primarily relies on the successfully accessed materials and supplementary searches conducted to fill contextual gaps where possible.

This review is structured as follows: Section 2 delves into the conceptualization of stress within the context of technological monitoring, examining foundational theories and the roles of affective computing, human–computer interaction, and computational empathy. Section 3 provides an overview of the IoT ecosystem for stress data acquisition, detailing wearable devices, sensor technologies, key physiological and behavioral biomarkers, and relevant standardized datasets. Section 4 focuses on the AI-driven analysis techniques applied to stress signals, covering data preprocessing, feature engineering, machine learning and deep learning models, and methods for real-time analysis. Section 5 highlights significant advancements in the field, including multimodal data fusion, personalization strategies, predictive modeling, and real-time intervention frameworks. Section 6 offers a critical assessment of the persistent challenges and limitations hindering progress, encompassing accuracy, generalizability, scalability, security, privacy, ethics, technological integration, and user experience. Section 7 outlines key future research directions aimed at addressing these challenges. Finally, Section 8 concludes the review by synthesizing the

main findings and reflecting on the transformative potential and responsible innovation required in this domain.

2. Conceptualizing Stress in the Technological Era

2.1. Foundational Stress Theories

Understanding how stress is conceptualized is fundamental to developing effective technological monitoring and intervention systems. Among the most influential frameworks is Lazarus and Folkman's Transactional Theory of Stress and Coping [7]. This theory posits that stress is not merely an external stimulus or an internal response but a dynamic transaction between an individual and their environment. Central to this transaction is the process of cognitive appraisal, which involves evaluating the significance of an encounter for one's well-being and assessing the resources available to cope with it [7].

The theory distinguishes between two key appraisal stages. Primary appraisal involves judging whether a situation is irrelevant, benign/positive, or stressful. If deemed stressful, it is further evaluated in terms of potential harm/loss, threat, or challenge. Secondary appraisal involves evaluating one's coping options and resources—such as social support, material assets, and personal coping skills—to manage the demands of the stressful situation [7]. Coping itself is defined as the constantly changing cognitive and behavioral efforts employed to manage these demands. Lazarus and Folkman initially identified two main coping functions: problem-focused coping (acting on the environment or oneself to change the source of stress) and emotion-focused coping (regulating the emotional distress associated with the situation) [7].

The relevance of this theory to AI-/IoT-based stress management is profound. It highlights that the stress experience is highly subjective and mediated by individual cognitive processes. A situation perceived as highly stressful by one person might be seen as a manageable challenge by another, depending on their appraisal. This implies that truly personalized and effective stress management systems should ideally move beyond merely detecting physiological stress responses. They should aim to understand or infer the individual's cognitive appraisal and typical coping patterns [7]. However, a significant disconnect exists between this theoretical understanding and current technological practice. Most contemporary IoT/AI systems excel at measuring the physiological outcomes of the stress response—changes in heart rate variability (HRV), electrodermal activity (EDA), skin temperature, and respiration. These are downstream effects triggered after the appraisal process has occurred. Few systems effectively capture or model the cognitive appraisal stage itself. This represents a critical gap: current systems can detect that someone is stressed but not why they perceive the situation as stressful based on their unique interpretation and resource evaluation. Bridging this gap necessitates integrating richer contextual information, leveraging ecological momentary assessments (EMAs) to capture subjective experiences in real time [1], or developing novel sensing modalities capable of inferring cognitive states more directly. Without addressing the appraisal component, interventions remain reactive to physiological symptoms rather than potentially targeting the cognitive roots of the stress response, limiting the depth of personalization and effectiveness.

2.2. Affective Computing, HCI, and Computational Empathy in Stress Management

The development of technology-mediated stress management systems intersects significantly with several key areas within computer science. Affective computing is the field dedicated to creating systems and devices that can recognize, interpret, process, and even simulate human affect or emotions [8]. This capability is central to stress detection systems, which aim to infer an internal emotional state (stress) from observable physiological or behavioral signals. Research in affective computing covers foundational

emotion theories, signal acquisition, sentiment analysis, multimodal fusion, and emotion generation [8]. Human–computer interaction (HCI) provides the principles and methodologies for designing systems that are usable, acceptable, effective, and provide a positive user experience. For stress monitoring technologies, which often involve intimate data collection and personal interventions, user-centered design is paramount. Issues such as the comfort and aesthetics of wearable devices, the intuitiveness of interfaces, the clarity of feedback, and the perceived burden of interaction are crucial to user adoption and long-term engagement [9].

Emerging from these fields is the concept of computational empathy, which refers to the ability of computational systems to perceive, understand, and respond appropriately to human emotional states [10]. In the context of stress management, an “empathic” system might not only detect high stress levels but also offer supportive feedback or tailored interventions in a sensitive manner [8]. The potential benefits include more natural and supportive human–machine interactions, particularly in areas like mental health support, where AI chatbots are increasingly used [11]. However, achieving genuine computational empathy faces significant challenges. While AI can learn to recognize patterns associated with stress [12] and generate responses that mimic empathy [11], these systems lack genuine consciousness, understanding, or the shared experience that underpins human empathy. There is a risk that simulated empathy, if not carefully designed and transparently communicated, could be perceived by users as inauthentic, manipulative, or even unsettling (falling into an “uncanny valley” of emotional interaction), potentially eroding trust and hindering the system’s therapeutic goals [11]. Therefore, the development of stress management systems requires the careful integration of accurate affective state detection (affective computing), a thoughtful interaction design prioritizing user needs and experience (HCI), and a cautious, transparent approach to implementing empathic responses (computational empathy) that acknowledges the system’s limitations while maximizing its supportive potential [9].

3. IoT Ecosystem for Stress Data Acquisition

The foundation of any AI-driven stress monitoring system lies in the reliable acquisition of relevant data. The IoT provides the infrastructure for this, connecting various sensors capable of capturing physiological and behavioral signals indicative of stress. This section provides an overview of the available IoT sensors suitable for stress detection, which are summarized in Table 1.

3.1. Wearable Devices and Sensor Technologies

Wearable technology is the cornerstone of continuous, ambulatory stress monitoring. Devices come in various form factors, each with associated trade-offs:

- **Wristbands/Smartwatches:** The most common form factor, offering convenience and social acceptability. They typically incorporate PPG, EDA, temperature sensors, and accelerometers. Signal quality, particularly for PPG-derived HRV, can be susceptible to motion artifacts [13].
- **Chest Straps/Patches:** They often provide higher-fidelity signals, especially for ECG, due to stable placement closer to the heart [14]. They may also include respiration sensors (impedance pneumography) and accelerometers. They can be less comfortable for long-term wear compared with wristbands.
- **Rings:** They offer a discreet alternative, often focusing on PPG, temperature and activity tracking.
- **Earbuds/Headsets:** They can potentially integrate EEG sensors (ear-EEG) or PPG.
- **Smart Clothing:** Textiles with embedded sensors (e.g., ECG electrodes [5]) offer potential for seamless integration but face challenges in durability and washability.

A key trend is the integration of multiple sensors into single devices to enable multimodal data collection [15]. The choice of device involves balancing factors like user comfort, desired signal quality, battery life, cost, and the specific physiological parameters of interest [4]. Beyond wearables, specific contexts may utilize other IoT sensors. Examples include thermal cameras integrated into vehicles to monitor driver facial temperature changes associated with stress [5]; Force Sensitive Resistors (FSRs) embedded in chair seats to classify sitting posture, which can correlate with fatigue or discomfort [16]; and even sensors embedded within computer mice to capture physiological data (PPG and GSR) during office work [17].

3.2. Key Physiological Biomarkers and Sensors

Several physiological signals, primarily governed by the Autonomic Nervous System (ANS), are commonly monitored as indicators of stress:

- **Electrocardiogram (ECG):** Measures the heart's electrical activity via electrodes placed on the skin. It is considered the gold standard for deriving heart rate variability (HRV), a sensitive indicator of ANS balance and stress. ECG sensors are often found in chest straps or require specific finger/hand placement on devices [14].
- **Photoplethysmography (PPG):** An optical technique measuring changes in blood volume in the microvascular bed of tissue, typically at the wrist or finger. It provides Blood Volume Pulse (BVP) from which HR and HRV can be estimated. While convenient for wrist-worn devices, PPG-derived HRV is generally considered less accurate than ECG-derived HRV, especially during movement [12].
- **Electrodermal Activity (EDA)/Galvanic Skin Response (GSR):** It measures changes in the electrical conductance of the skin driven by sweat gland activity controlled by the sympathetic nervous system (the "fight or flight" branch of the ANS). Increased EDA/GSR typically indicates heightened arousal or stress [1]. Sensors are commonly placed on the wrist or fingers/palm. EDA is sensitive to motion artifacts and environmental factors like temperature and humidity [13].
- **Skin Temperature (SKT):** Peripheral skin temperature can decrease during acute stress due to vasoconstriction (blood vessels narrowing) directed by the sympathetic nervous system. It can be measured by thermistors placed on the skin, often on the wrist [12]. Facial skin temperature changes have also been explored using thermal cameras [5].
- **Respiration (RSP):** Stress often leads to faster, shallower breathing. Respiration rate and pattern can be measured directly using chest straps (measuring chest expansion/contraction via impedance or strain gauges) or indirectly estimated from ECG or PPG signals [12].
- **Electromyogram (EMG):** It measures the electrical activity produced by skeletal muscles. Increased muscle tension (e.g., in the shoulders, neck, or jaw) can be a physical manifestation of stress. EMG sensors require electrode placement over the muscle of interest. It is less commonly used in general-purpose stress wearables compared with ANS indicators.
- **Electroencephalogram (EEG):** It records the electrical activity from the brain via electrodes placed on the scalp (or potentially in/around the ear). EEG provides direct insights into brain states, including cognitive load, relaxation (alpha waves), and affective responses, making it highly relevant for stress research [18]. However, traditional EEG requires cumbersome electrode caps, limiting its use for continuous monitoring in daily life. Wearable EEG solutions are emerging but still face usability challenges [19]. EEG is often employed in laboratory studies or specific applications like neurofeedback [20].

- **Accelerometer (ACC)/Motion Data:** Inertial sensors measuring acceleration (and often rotation via gyroscopes) are ubiquitous in wearables. While not direct stress measures, they are crucial to quantifying physical activity levels. This is vital because physical exertion causes changes in HR, respiration, and temperature that can mimic or mask stress responses. Motion data are, therefore, essential to contextualizing physiological signals, filtering out activity-related noise, or distinguishing mental stress from physical load [12].

An inherent tension exists in the choice of sensors. Peripheral sensors monitoring ANS activity (PPG, EDA, SKT, and RSP via chest straps) are generally more convenient and suitable for continuous, unobtrusive monitoring in daily life using common form factors like wristbands or patches. However, these signals are indirect indicators of stress and can be significantly confounded by non-stress factors like physical activity, ambient temperature changes, or even posture [12]. Conversely, EEG provides a more direct measure of brain activity related to cognitive and affective processes underlying stress [21]. Yet, traditional EEG setups are impractical for everyday wear, limiting their ecological validity [19]. This trade-off between the convenience and ecological validity of peripheral sensing and the directness and richness of central nervous system sensing (EEG) drives research in two main directions: (1) multimodal fusion, combining data from multiple sensor types (e.g., peripheral + motion, or peripheral + EEG when feasible) to achieve a more comprehensive and robust assessment [22], and (2) the development of sophisticated signal processing and machine learning algorithms capable of disentangling the stress-related components from confounding factors within the data streams from convenient peripheral sensors [13].

Table 1. Overview of common IoT sensors for stress detection.

Sensor Type	Signal Measured	Common Location(s)	Key Advantages	Key Limitations/Challenges	
ECG	Electrical Activity of the Heart	Chest and Fingers/Hands	Gold standard for HRV; high accuracy	Often requires specific placement (chest strap); can be obtrusive	[14]
PPG	Blood Volume Pulse (BVP)	Wrists and Fingers	Convenient for wearables (wristbands and rings); estimates HR and HRV	Susceptible to motion artifacts; less accurate HRV than ECG, especially during movement	[12]
EDA/GSR	Skin Conductance (sweat gland activity)	Wrist, Fingers, and Palms	Reflects sympathetic nervous system arousal; widely used	Sensitive to motion, temperature, and humidity; individual variability	[1,13]
SKT	Skin Temperature	Wrists and Forehead (thermal cam)	Non-invasive; can indicate peripheral blood flow changes	Influenced by ambient temperature; response can be slow or variable	[12,19]
RSP	Respiration Rate/Pattern	Chest (strap) and derived (ECG/PPG)	Direct measure of breathing changes linked to stress	Chest straps can be uncomfortable; derived methods may be less accurate	[12]
EMG	Muscle Electrical Activity/Tension	Various muscle groups	Directly measures muscle tension associated with stress	Requires specific electrode placement over muscles; less common for general stress	[12]
EEG	Brain Electrical Activity	Scalp and Ears (ear-EEG)	Direct measure of brain states (cognitive/affective); high temporal resolution	Traditionally requires cumbersome setup; wearable options still evolving; susceptible to artifacts (muscle and eye movement)	[18–21]

Table 1. Cont.

Sensor Type	Signal Measured	Common Location(s)	Key Advantages	Key Limitations/Challenges	
ACC	Acceleration/Motion	Wrists, Chest, Waist, etc.	Quantifies physical activity; essential context for physiological signals	Does not directly measure stress; primarily used for context/artifact removal	[12]
Camera	Facial Images/Video	External	Captures facial expressions linked to emotion/stress	Privacy concerns; requires line of sight; lighting variations; computationally intensive processing	[23]
FSR	Force/Pressure	Chair seat and insoles	Can detect posture changes or pressure points related to discomfort/stress	Context-specific (e.g., sitting posture); indirect measure	[16]
Thermal Cam	Infrared Radiation (Surface Temperature)	External	Non-contact temperature measurement (e.g., facial temperature patterns)	Sensitive to environmental temperature, distance, and emissivity; can be expensive; privacy concerns	[5]

3.3. Behavioral Indicators

Beyond physiological signals, AI and the IoT enable the monitoring of behavioral patterns that can indicate stress:

- **Facial Expressions:** Computer vision algorithms, particularly Convolutional Neural Networks (CNNs), can analyze images or video streams to detect facial muscle movements corresponding to basic emotions (e.g., anger, fear, sadness, happiness, and surprise) or specific Action Units (AUs) linked to stress [23]. IoT-enabled cameras, potentially integrated into smart environments or personal devices, facilitate this data capture. Research in this area is rapidly advancing to address real-world challenges. For instance, a work on “Facial Expression Recognition with Visual Transformers and Attentional Selective Fusion” [24] proposes a method (VTFF) that converts facial images into sequences of “visual words” and uses self-attention mechanisms to model the relationships between them, achieving high accuracy on “in-the-wild” datasets like RAF-DB and FERPlus. Such approaches are particularly promising for stress monitoring systems because they are designed to be robust to occlusions, varying head poses, and complex backgrounds, which are common outside of laboratory settings. However, continuous camera monitoring raises significant privacy implications [25].
- **Posture:** Changes in sitting or standing posture, such as slouching or increased rigidity, can be associated with stress, fatigue, or negative affect. Sensors like FSRs in chairs [16] or wearable inertial sensors can potentially track posture over time.
- **Physical Activity Levels:** As measured by accelerometers, deviations from typical activity patterns (e.g., unusual restlessness or lethargy) might correlate with stress levels. Activity data also provide crucial context for interpreting physiological signals [12].
- **Voice Characteristics:** Although not extensively detailed for stress in the accessible sources, affective computing research indicates that vocal parameters like pitch, intensity, speech rate, and jitter/shimmer can change with the emotional state, including stress [8]. This could be captured via microphones in smartphones or other IoT devices.
- **Computer Interaction Patterns:** In office or work-from-home settings, changes in typing speed, error rates, mouse movement patterns, or application usage could potentially serve as subtle behavioral indicators of stress or cognitive load [8].

3.4. Standardized Datasets for Research

The development and validation of AI models for stress detection heavily rely on the availability of high-quality, annotated datasets. In Table 2 we summarize main aspects of publicly available datasets which have become benchmarks in the field:

- **WESAD (Wearable Stress and Affect Detection):** A widely cited multimodal dataset collected from 15 participants during a laboratory study [26]. It includes physiological signals from both chest-worn (ECG, EDA, EMG, respiration, temperature, and three-axis acceleration at 700 Hz) and wrist-worn (BVP at 64 Hz, EDA at 4 Hz, temperature at 4 Hz, and three-axis acceleration at 32 Hz) devices. The protocol involved baseline (neutral), stress induction (using elements similar to the Trier Social Stress Test), and amusement phases. Self-report questionnaires (e.g., PANAS and STAI) provide subjective labels [26]. WESAD facilitates research comparing sensor locations and modalities for classifying baseline, stress, and amusement states [15,24].
- **AMIGOS (A Dataset for Affect, Personality and Mood Research on Individuals and GroupS):** This dataset focuses on affect elicited by short and long emotional videos viewed by 40 participants, both individually and in small groups [27]. It includes physiological data from wearable sensors (EEG, ECG, and GSR) and video recordings (frontal, full-body RGB, and depth). A key feature is the inclusion of personality assessments (Big Five Inventory) and mood assessments (PANAS), allowing researchers to investigate the interplay of personality, mood, social context, and affective responses [27]. Affect is annotated via self-assessment (valence, arousal, dominance, liking, familiarity, and basic emotions) and external assessment (valence and arousal) [27].
- **SWEET (Stress in Work EnvironmEnT):** Referenced as the data source in studies focusing on real-world stress detection, this dataset is significant because it was collected in a free-living environment over five consecutive days from 240 volunteers. Data include ECG, skin conductance (SC), skin temperature (ST), and accelerometer (ACC) signals from wearable patches and wristbands. Stress levels were self-reported periodically via questionnaires [3]. SWEET is crucial to developing and testing models intended for deployment outside controlled laboratory settings.

Table 2. Comparative analysis of major public datasets for stress detection.

Dataset	# Subjects	Modalities	Scenario	Key Features and Limitations
WESAD	15	Physiological: ECG, EDA, EMG, RESP, TEMP, and ACC (chest); BVP, EDA, TEMP, and ACC (wrist). Subjective: Questionnaires (PANAS and STAI).	Lab	Features: High-res multimodal data from two locations; includes three affective states (baseline, stress, and amusement). Limitations: Very small sample size; controlled lab setting limits ecological validity.
AMIGOS	40	Physiological: EEG, ECG, and GSR. Behavioral: Frontal video and full-body video (RGB and depth). Subjective/Trait: Self-assessment (valence and arousal), personality (Big Five), and mood (PANAS).	Lab	Features: Includes personality/mood data for individual differences; explores social context (solo/group viewing). Limitations: Relatively small sample size; lab setting.
SWEET	>1000 (overall) 240 (cited cohort)	Physiological: ECG, SC, ST, and ACC (patches and wristbands). Contextual: GPS, phone activity, and noise level. Subjective: Periodic questionnaires (EMA).	In the wild	Features: Large-scale, real-world data over 5 days; includes smartphone context; high ecological validity. Limitations: Stress labels based only on periodic self-reports, which can be less precise.

These standardized datasets are invaluable for promoting reproducibility, allowing researchers to benchmark algorithms, and facilitating comparisons across studies. However, they are not without limitations. Many publicly available datasets suffer from relatively

small sample sizes (often under 50 participants), potentially limiting statistical power and the generalizability of findings. Most were collected under controlled laboratory conditions, which may not accurately reflect the complexities and variability of real-life stressors and environments, raising concerns about ecological validity. Datasets may also exhibit demographic biases (e.g., skewed towards younger adults or specific genders), further impacting generalizability [19]. Furthermore, inconsistencies in experimental protocols for stress induction and varying methods for assigning ground-truth stress labels (self-report, observer ratings, and task performance) make direct comparisons between studies challenging. The need for larger, more diverse datasets collected in naturalistic settings, potentially utilizing EMAs for labeling, remains a critical requirement for advancing the field [13].

4. AI-Driven Analysis of Stress Signals

In this section we will compare different approaches to data analysis and detail which kinds of data are more suitable for each one. Key aspects of the studied processing methods will be summarized in Table 3. Before processing, data has to be acquired via IoT sensors and here AI techniques are also employed to transform raw signals into meaningful insights into stress levels. This involves several stages, from preparing the data to applying sophisticated learning algorithms, as depicted in Figure 1.

4.1. Data Preprocessing and Feature Extraction and Selection

Raw sensor data are often noisy and require significant preprocessing before analysis. Common steps include the following:

- **Noise Filtering:** The application of digital filters to remove unwanted noise, such as baseline drift or high-frequency interference. Examples include median filters and Butterworth low-pass filters for ECG [28], Kalman filtering for general noise reduction [29], and Wavelet Packet Transform (WPT) for signal enhancement and noise minimization in EEG [21].
- **Artifact Removal/Mitigation:** The identification and correction or removal of segments of data corrupted by artifacts, particularly motion artifacts, which heavily affect signals like PPG and EDA. Techniques might involve signal quality indices, interpolation, or algorithms specifically designed to separate physiological signals from motion noise.
- **Signal Segmentation:** The division of continuous data streams into smaller, manageable windows (e.g., 1 min, 5 min, or longer [13]) for feature extraction and analysis. The choice of window size can impact the results [13].
- **Normalization/Standardization:** The scaling of data to a common range (e.g., 0 to 1 or with zero mean and unit variance) to prevent features with larger values from dominating the learning process [13].

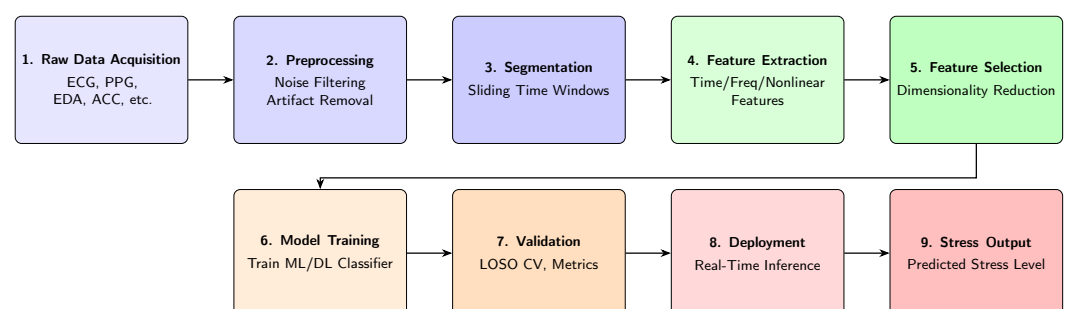


Figure 1. General data processing and analysis pipeline for real-time stress detection. The system processes raw multimodal sensor data through preprocessing, segmentation, feature extraction, model training and validation, culminating in real-time stress level inference.

Following preprocessing, relevant information needs to be extracted in the form of features. Feature engineering is the calculation of numerical representations (features) that capture characteristics of the signals relevant to stress. This includes the following:

- Time-domain features: Statistical measures like mean, standard deviation (SD), variance, median, min/max, zero-crossing rate, skewness, kurtosis, line length, and nonlinear energy calculated over signal windows [13].
- Frequency-domain features: The analysis of the power distribution across different frequency bands. For HRV, this includes power in low-frequency (LF) and high-frequency (HF) bands, and the LF/HF ratio, which reflect sympathetic and parasympathetic activity [12]. Power Spectral Density (PSD) analysis is common for EEG and HRV [21]. Instantaneous wavelet mean/band frequency are also used [21].
- Nonlinear features: It measures capturing complexity or chaotic dynamics, such as entropy (e.g., Shannon Entropy [21]), fractal dimensions, or recurrence quantification analysis.
- Domain-specific features: Metrics specific to certain signals, e.g., number and amplitude of skin conductance responses (SCRs) for EDA, or various time- and frequency-domain HRV metrics like RMSSD (Root Mean Square of Successive Differences) and SDNN (standard deviation of NN intervals) [12]. Hjorth parameters (activity, mobility, and complexity) are used for EEG [21]. Encoding techniques like Local Binary Patterns (LBPs) might be applied to time series [21].

Given the potentially large number of extracted features, feature selection or dimensionality reduction is often necessary. Feature selection/reduction aims to identify the most informative subset of features, reducing model complexity, improving generalization, and potentially enhancing interpretability. Techniques include the following:

- Filter methods: The selection of features based on statistical properties (e.g., correlation with the target variable, mutual information, Chi-square test, *t*-test, and Minimum Redundancy Maximum Relevance—mRMR) independent of the learning algorithm.
- Wrapper methods: The use of the performance of a specific learning algorithm to evaluate feature subsets.
- Embedded methods: Feature selection integrated within the learning algorithm (e.g., LASSO regression).
- Dimensionality reduction: The transformation of features into a lower-dimensional space while preserving variance, using methods like Principal Component Analysis (PCA) or Independent Component Analysis (ICA). Optimization algorithms like the Archimedes Optimization Algorithm (AOA) combined with the Analytical Hierarchical Process (AHP) have also been proposed for feature selection [21].

Finally, practical data issues must be addressed. Handling missing data: Strategies like imputation (e.g., replacing missing values with mean and median or using more sophisticated methods like Kalman filters) are needed, as missing data are common in real-world wearable sensor streams [4]. Class imbalance: Stress datasets often have significantly more non-stress samples than stress samples. This imbalance can bias classifiers towards the majority class. Techniques to address this include the following:

- Resampling: The under-sampling of the majority class, the over-sampling of the minority class, or the use of synthetic data generation methods like SMOTE (Synthetic Minority Over-sampling Technique) or ADASYN (Adaptive Synthetic Sampling). However, resampling can lead to information loss (under-sampling) or overfitting (over-sampling, SMOTE) [13].

Table 3. Comparative analysis of AI/ML algorithms for stress detection.

Algorithm	Category	Input Data Modality	Reported Accuracy	Context (Lab, Real-World, or Dataset)	Noted Strengths	Noted Weaknesses/Challenges	
SVM	Supervised ML	HR, PPG, Skin Response, EEG, and Physiological	93–96%	Lab and Real-World (SWEET), Various	Often high accuracy; effective in high dimensions	Can be sensitive to parameter choice; less interpretable than trees	[1]
Random Forest (RF)	Ensemble ML	HRV, EDA, and Physiological	98.3%	Lab and Real-World (SWEET and Stress-Lysis)	Robust to noise/overfitting; good performance; feature importance	Can become complex with many trees; less interpretable than single DT	[1,13]
k-NN	Supervised ML	Physiological	95.7%	Lab	Simple to implement; non-parametric	Sensitive to feature scaling; computationally expensive at prediction time	[12,13]
XGBoost	Ensemble ML (Boosting)	Physiological	98.98%	Real-World (SWEET)	High accuracy; regularization; handles missing data	Can be complex to tune; less interpretable	[13,14]
Stacked Ensemble	Ensemble ML	Temp, Humidity, and Steps	99.5%	Lab (Stress-Lysis dataset)	Potentially the highest accuracy by combining models	Increased complexity; potential for overfitting meta-model	[30]
CNN	Deep Learning	HRV, ECG, EEG, and Facial Expr.	92.8%	Lab, Various	Automatic feature learning; good for spatial patterns	Requires many data; black box; computationally intensive	[23,31]
LSTM/Bi-LSTM	Deep Learning (RNN)	Physiological, Time Series, and ECG/BP	98.86%	Lab, Various	Captures temporal dependencies; good for sequences	Requires many data; can be slow to train; black box	[29]

Note: The reported performance metrics are highly context-dependent (dataset, features, and validation method) and should be interpreted cautiously.

4.2. Machine Learning Techniques for Classification and Prediction

Supervised machine learning algorithms are widely used to classify stress levels based on the extracted features. The task is typically framed as either binary classification (e.g., stress vs. no stress/baseline) or multi-class classification (e.g., no stress, low stress, and high stress; or baseline vs. stress vs. amusement, as in WESAD [26]). Commonly employed algorithms include the following:

- **Support Vector Machines (SVMs):** A powerful algorithm that finds an optimal hyperplane to separate different classes. SVMs have been frequently used and often demonstrate high accuracy in stress detection tasks across various data types [1].
- **Random Forest (RF):** An ensemble method based on multiple decision trees. RF is robust to noise, less prone to overfitting than single decision trees, and often achieves high performance. It was found to be among the most frequently used and best-performing algorithms in several reviews and studies [1].
- **K-Nearest Neighbors (k-NN):** A simple, non-parametric algorithm that classifies a sample based on the majority class of its k-nearest neighbors in the feature space. Its performance can be sensitive to the choice of k and the distance metric [12].
- **Decision Trees (DTs):** Tree-like models where internal nodes represent feature tests and leaf nodes represent class labels. They are prone to overfitting if not pruned properly [3].
- **Naive Bayes (NB):** A probabilistic classifier based on Bayes' theorem with strong (naive) independence assumptions between features. It is often simple and efficient [1].
- **Logistic Regression (LR):** A linear model used for binary classification tasks [1].
- **Boosting Algorithms:** Ensemble methods that build models sequentially, with each new model focusing on correcting the errors of the previous ones. Examples include AdaBoost [12] and Gradient Boosting Machines (GBMs), including XGBoost [14], which often yield state-of-the-art results on tabular data.
- **Ensemble Methods:** Combining predictions from multiple individual models (base learners) often leads to improved accuracy and robustness compared with single models. RF and Gradient Boosting are examples of ensemble techniques. Stacked Ensembles, where the predictions of base models (e.g., RF and GB) are used as input features for a meta-model, can further enhance performance by leveraging the diversity of the base learners. One study reported an accuracy of 99.5% using a Stacked Ensemble on a specific dataset, though such high scores often warrant scrutiny regarding the dataset characteristics, evaluation methodology, and potential for overfitting [30].

Validation Techniques: Rigorous evaluation is crucial to assessing model performance and generalizability. Common techniques include the following:

- **K-Fold Cross-Validation:** The dataset is divided into k subsets (folds). The model is trained on $k - 1$ folds and tested on the remaining fold, a process which is repeated k times. This provides a more robust estimate of performance than a single train-test split [1].
- **Leave-One-Subject-Out (LOSO) Cross-Validation:** Particularly relevant for personalized models or assessing inter-subject generalizability. The model is trained on data from all subjects except one, tested on the left-out subject, and repeated for each subject. Performance is typically measured using metrics like accuracy, precision, recall (sensitivity), specificity, F1-score, and Area Under the ROC Curve (AUC). Testing models on completely independent datasets (recorded under different conditions or from different populations) is considered the strongest form of validation for assessing true generalizability, but this is rarely performed in practice [13].

4.3. Deep Learning Models in Stress Research

Deep learning (DL) models, particularly neural networks with multiple layers, have gained traction in stress research due to their ability to automatically learn complex patterns and hierarchical features directly from raw or minimally processed data. This can potentially reduce the need for extensive manual feature engineering, which requires significant domain expertise. Key DL architectures applied to stress detection include the following:

- Convolutional Neural Networks (CNNs): Primarily designed for grid-like data such as images, CNNs excel at extracting spatial hierarchies of features. In stress detection, they are applied
 - To spectrograms or other 2D representations of physiological time series (e.g., ECG and EEG).
 - Directly to 1D time-series data to learn local patterns and features.
 - In facial expression recognition from images or video frames [23].
- Recurrent Neural Networks (RNNs): They are designed to handle sequential data, which makes them suitable for modeling time series like physiological signals, and include the following:
 - Long Short-Term Memory (LSTM) Networks: A type of RNN specifically designed to capture long-range temporal dependencies, mitigating the vanishing gradient problem of simple RNNs. LSTM networks are frequently used for analyzing ECG, EEG, and other physiological time series.
 - Bidirectional LSTM networks (Bi-LSTM networks): The process sequences in both forward and backward directions, allowing the model to utilize past and future context within the sequence, potentially improving performance [29]. A study using a Fuzzy Inference System (FIS) combined with Bi-LSTM (FBiLSTM) reported high accuracy (98.86%) in heart disease prediction, suggesting potential for stress-related applications [29].
- Transformers: Originally developed for natural language processing, Transformer models use self-attention mechanisms to capture dependencies regardless of their distance in the sequence. Their application in physiological signal analysis, including stress detection, is an emerging area [19].
- Hybrid Models: The combination of different architectures to leverage their respective strengths. For example, CNNs can be used for initial feature extraction from segments of the signal, followed by LSTM to model the temporal relationships between these features (e.g., DCNN-LSTM [21] as depicted in Figure 2). Autoencoders, another type of neural network, can be used for unsupervised feature learning or dimensionality reduction.

Despite their potential, DL models present challenges. They typically require a large number of labeled data for effective training, which can be scarce in the context of stress research, especially for personalized models. DL models can also be computationally intensive to train and deploy, particularly on resource-constrained wearable devices [32]. Perhaps the most significant limitation is their lack of interpretability—they often function as “black boxes,” making it difficult to understand why they make a particular prediction. This lack of transparency can be a major barrier to trust and adoption, especially in clinical settings [19].

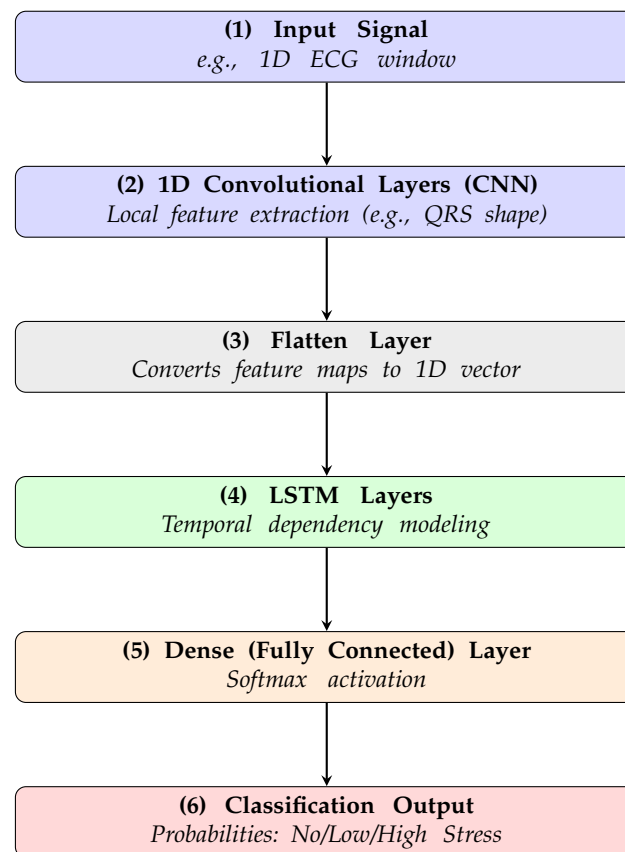


Figure 2. Hybrid CNN-LSTM architecture for stress detection from physiological signals. CNN layers extract local features (e.g., QRS morphology), which are temporally modeled by LSTM layers. Final classification is performed via a dense layer with softmax activation.

4.4. Techniques for Real-Time Analysis and Complex Event Processing (CEP)

Implementing stress detection systems that operate in real time imposes specific constraints. To provide a precise definition of “real time”, it is essential to distinguish between different system latencies depending on the application.

Hard real time (sub-second latency) is necessary for biofeedback interventions that require near-instantaneous response. *Soft real time* (second-to-minute latency) is suitable for just-in-time adaptive interventions (JITAI), where suggesting a break or a breathing exercise can tolerate a brief delay. *Near-real time* or batch processing (minutes to hours) can suffice for generating daily or weekly summaries of stress trends. Most current research methods, which use analysis windows of several minutes and complex feature calculations, introduce inherent delays that place them in the soft- or near-real-time category. This necessitates efficient algorithms and system architectures:

- **Sliding Window Analysis:** Data are typically processed in overlapping time windows, allowing for continuous monitoring as new data arrive. The size and overlap of these windows are important parameters affecting responsiveness and computational load.
- **Complex Event Processing (CEP):** While not explicitly detailed in the accessible sources for stress detection (e.g., ref. [33] was inaccessible), CEP engines are designed to analyze streams of data (events) from multiple sources, identify complex patterns or sequences of events over time, and trigger actions based on these patterns. This paradigm seems highly relevant for stress monitoring, potentially enabling the detection of not just instantaneous stress levels but also patterns like “stress pileup” (multiple stressors occurring in close succession without adequate recovery) [34] or specific sequences of physiological changes leading to a stress event [3].

- **Edge/Fog Computing:** To address latency issues associated with sending a large number of sensor data to the cloud for processing, edge or fog computing architectures are being explored. In this model, some data preprocessing and analysis occur closer to the data source (e.g., on the wearable device itself, a smartphone, or a local gateway) before potentially sending summarized results or critical events to the cloud. This can reduce bandwidth requirements, improve response times, and enhance privacy by keeping raw data local [4].

However, the term “real time” itself often lacks precise definition in the literature. Some studies may refer to near-instantaneous feedback loops (sub-second), while others might consider processing within minutes or even providing daily summaries as “real-time” monitoring. This ambiguity arises because the required latency depends heavily on the application. For instance, biofeedback interventions might necessitate sub-second updates, whereas suggesting a break based on rising stress trends might tolerate delays of several minutes. The processing pipelines described often involve windowing over several minutes [13], complex feature extraction, or cloud-based computations [30], all of which introduce inherent delays. Truly instantaneous processing might require simpler models running directly on resource-constrained devices [4], potentially sacrificing some accuracy or complexity compared with cloud-based deep learning approaches [29]. Future work should strive for clearer definitions of “real time” based on the specific latency requirements of the intended stress detection or intervention task.

4.5. Architectural Considerations: Transformers vs. Recurrent Models for Physiological Time Series

To address the need for greater technical depth, it is crucial to analyze the architectural debate between Transformer models and recurrent models like LSTM for physiological time-series analysis [35,36].

LSTM models process data sequentially, making them inherently suited for capturing temporal dependencies in time-series data. Their gating mechanism allows them to maintain long-term information, which is vital to understanding physiological dynamics. Having fewer parameters than Transformers, LSTM models can generalize better with limited training data and are computationally less demanding, which makes them more suitable for online learning and deployment on resource-constrained devices (edge computing).

Transformer models, on the other hand, use self-attention mechanisms to process all data points in a sequence in parallel. This allows them to capture global long-range dependencies more effectively than LSTM and significantly speeds up training. Their scalability makes them ideal for large datasets and complex models, and they have demonstrated state-of-the-art performance in ECG signal classification.

This architectural choice has direct implications for the overall architecture of the stress detection system and its real-time capabilities. A decision to use an LSTM model might favor an edge-centric architecture, where processing is performed on the user’s device, prioritizing low latency and data privacy. Conversely, choosing a more powerful but resource-intensive Transformer model would likely necessitate a cloud-based architecture, where data are sent for analysis, introducing higher latency and potential privacy concerns but allowing for more complex models. Hybrid models, such as CNN–Transformer–LSTM architectures, are emerging, attempting to combine the strengths of each approach: local feature extraction from CNNs, global temporal relationship understanding from Transformers, and long-term sequential dependency capture from LSTM models. The selection of the optimal architecture, therefore, is not just a matter of model performance but also a fundamental design decision that balances accuracy, latency, computational cost, and privacy.

5. Advancements in Real-Time Stress Detection and Modulation

The field is continually evolving beyond basic stress classification, with significant advancements in integrating multiple data sources, personalizing models, predicting future states, and developing closed-loop intervention systems. These new features are pivotal to achieving true real-time assessment and intervention over user stress levels. This section will give deeper context on these new aspects.

5.1. Multimodal Data Fusion Strategies

Recognizing that stress manifests across physiological, behavioral, and subjective domains, researchers increasingly employ multimodal approaches, fusing data from different sources to achieve more comprehensive and robust stress assessment [32]. The rationale is that different modalities capture complementary information; for example, physiological signals might indicate arousal, while facial expressions or the voice tone provides valence information, and motion data offer context [1]. Combining these streams can potentially improve accuracy, enhance robustness against noise or limitations in a single modality, and provide a richer understanding of the individual's state [32].

Fusion can occur at different stages [32]:

- Early Fusion (Feature Level): Features extracted from different modalities are concatenated into a single vector before being fed into a classifier.
- Intermediate Fusion: Features from different modalities are combined at intermediate layers within a model (e.g., in a deep neural network).
- Late Fusion (Decision Level): Separate classifiers are trained for each modality, and their outputs (e.g., class probabilities or decisions) are combined using methods like averaging and voting, or a meta-classifier.
- Hybrid Fusion: Combines multiple fusion strategies.

Examples from the literature include fusing physiological signals (e.g., ECG and EDA) with motion data from accelerometers to differentiate mental stress from physical exertion, integrating physiological data with electronic health records (EHRs) for cardiovascular risk prediction [32], combining facial expression features with speech features by using attention mechanisms [37], and fusing EEG with other physiological signals for improved emotion recognition accuracy [22]. While promising, multimodal fusion introduces challenges related to data heterogeneity (different sampling rates and formats), the temporal alignment of data streams, the determination of the optimal fusion strategy, the handling of missing data in one or more modalities, and the management of the increased complexity of the resulting models [32].

5.2. Personalization Approaches to Stress Modeling

A one-size-fits-all approach to stress detection is often inadequate due to significant inter-individual variability. People differ in their baseline physiological levels, the magnitude and pattern of their physiological responses to stressors, their perception of what constitutes a stressor (appraisal), and their coping mechanisms. Consequently, generic models trained on group data frequently exhibit poor performance when applied to new individuals. Personalization is, therefore, crucial to developing accurate and effective stress monitoring systems [13].

Several strategies are employed to personalize stress models:

- User-Specific Calibration: Collecting baseline data from an individual under non-stress conditions to establish personalized thresholds or normalize subsequent measurements.
- Person-Specific Models: Training separate models for each individual by using their own collected data. This can yield high accuracy for that individual but requires sufficient labeled data collection per person, which can be burdensome [13].

- **Transfer Learning/Adaptive Algorithms:** Starting with a general model trained on a larger dataset and then fine-tuning or adapting it using a smaller number of data from the target individual.
- **Clustering-Based Personalization:** Grouping individuals with similar physiological response patterns (identified through unsupervised learning techniques like K-means clustering or Self-Organizing Maps—SOMs) and training group-specific models.
- **Adaptive Baselines:** Developing models that can dynamically adjust an individual's baseline physiological levels over time to account for factors like circadian rhythms, acclimatization, or changes in health status. Multi-task learning has been suggested as a way to refine algorithms that adapt to individual baselines [19].

A key challenge remains the acquisition of sufficient high-quality labeled data for robust personalization, especially for fully person-specific models [1]. Techniques like self-supervised learning (SSL), which can leverage a large number of unlabeled biosignal data, are being explored to pre-train models that can then be fine-tuned with limited labeled data [19].

5.3. Predictive Stress Modeling Capabilities

Beyond detecting stress as it occurs, there is growing interest in developing models that can predict the likelihood of future stress states or adverse events. The motivation is to enable proactive interventions that can prevent stress from escalating or mitigate its negative consequences before they fully manifest. For example, predicting an imminent high-stress episode could allow an individual to take preemptive actions, such as engaging in relaxation techniques or modifying their schedule.

Techniques used for predictive modeling often involve analyzing temporal trends in physiological data by using time-series forecasting methods, such as LSTM or other sequence models. These models learn patterns from past data to forecast future values or classify upcoming states. For instance, models might predict future heart rate variability patterns or directly forecast the probability of transitioning into a stressed state within a defined future time window (e.g., the next 5, 10, or 30 min).

However, predictive stress modeling faces significant hurdles. It requires robust, longitudinal datasets capturing stress dynamics over extended periods. The models themselves can be complex and computationally demanding. Furthermore, defining and validating “future stress” is challenging—predicting stress minutes ahead might be feasible based on physiological precursors, but predicting stress hours or days ahead likely requires incorporating contextual information about anticipated events or environmental factors. Current predictive models often focus on forecasting physiological trajectories based primarily on past physiological data. This approach is largely reactive to trends that have already begun. A truly preventative system, aiming to avert the stress response altogether or mitigate it at the earliest possible stage, would ideally need to anticipate stressors before they significantly impact physiology. This requires moving beyond purely physiological forecasting to integrate predictive contextual information—such as upcoming calendar appointments, anticipated workload, traffic conditions, or known personal triggers—with an assessment of the individual's current physiological and psychological vulnerability. Such systems would necessitate more sophisticated causal modeling and deeper integration of diverse data sources to enable proactive, rather than merely predictive, interventions.

5.4. Real-Time Intervention Frameworks

The ultimate goal of real-time stress monitoring is often to enable timely and personalized interventions to help individuals manage their stress levels effectively. Several frameworks for such closed-loop systems are emerging:

- **Just-in-Time Adaptive Interventions (JITAI):** This is a prominent framework where interventions are delivered dynamically, precisely when and where they are needed most [34]. In the context of stress, a JITAI system would use real-time data from sensors (and potentially self-reports via EMAs) to detect moments of elevated stress or risk (e.g., high reactivity, slow recovery, and stress pileup [34]). Upon detection, the system triggers the delivery of a brief, context-appropriate intervention, typically via a smartphone or wearable device. Interventions might include guided breathing exercises, mindfulness prompts, cognitive reappraisal techniques, suggestions for positive activities, or reminders of personal coping resources. The “adaptive” nature means that the type, timing, or intensity of the intervention can be tailored based on the individual’s current state, context, and past responses [34].
- **Neurofeedback Systems:** These systems provide users with real-time feedback about their own brain activity, typically measured via EEG, enabling them to learn volitional control over specific neural patterns associated with desired mental states [20]. For stress reduction, neurofeedback might train users to increase alpha brainwave activity (associated with relaxation) or modulate activity in specific brain regions involved in emotion regulation (e.g., prefrontal cortex) [20]. Reinforcement learning (RL) algorithms can be used to optimize the feedback strategy and personalize the training process [20]. While powerful, EEG-based neurofeedback typically requires more specialized equipment and setup than JITAI delivered via standard mobile devices [20].
- **Biofeedback Systems:** Similar to neurofeedback, but using feedback from peripheral physiological signals like HRV, EDA, or respiration. For example, a system might visualize the user’s breathing pattern or HRV coherence in real time and guide them towards slower, deeper breathing to increase parasympathetic activity and promote relaxation.
- **Adaptive Environmental Adjustments:** Future systems might be integrated with smart environments to automatically adjust factors like lighting, ambient noise, or music based on the user’s detected stress level, creating a more calming atmosphere [8].

Developing effective real-time intervention systems involves challenges beyond accurate stress detection. Interventions must be designed to be engaging, brief enough to be used “in the moment,” and demonstrably effective [34]. Ensuring interventions are delivered appropriately and are not perceived as intrusive or burdensome is crucial to user acceptance and adherence [34]. Rigorous evaluation methodologies, such as randomized controlled trials (RCTs) using experimental medicine approaches, are needed to validate the efficacy of these closed-loop systems in reducing stress and improving health outcomes [34].

6. Critical Assessment: Persistent Challenges and Limitations

Despite significant progress, the widespread adoption and clinical translation of IoT- and AI-based real-time stress monitoring and modulation systems are hindered by several persistent challenges and limitations. This section will focus on the current challenges preventing further development of the stress detection techniques and propose possible solutions. Table 4 condenses the main aspects of the challenges described here.

6.1. Accuracy, Reliability, and Validity in Diverse Contexts

A major concern is the discrepancy often observed between the high accuracy figures reported in studies conducted under controlled laboratory conditions and the typically lower, more variable performance achieved when these systems are deployed in real-world, free-living environments [12]. Laboratory settings allow for standardized stress induction protocols, the minimization of confounding factors, and optimal sensor

placement, conditions rarely met in daily life. Real-world accuracy is compromised by numerous factors:

- **Motion Artifacts:** Physical movement can severely distort physiological signals, particularly PPG and EDA collected from the wrist [12].
- **Environmental Noise:** Ambient temperature, humidity, and noise can influence physiological readings or sensor performance [14,38].
- **Confounding Factors:** Physiological responses measured by sensors (e.g., increased heart rate and sweating) are not unique to psychological stress. They can also be caused by physical activity, illness, caffeine consumption, medication, or even positive excitement [12]. Disentangling stress-related changes from these confounders remains a major challenge [12].
- **Sensor Placement and Contact:** Variability in how users wear devices can affect signal quality [13].

Furthermore, establishing a reliable “ground truth” for stress against which to train and validate models is inherently difficult, especially in real time [12]. Stress is a subjective experience, and relying solely on self-reports introduces its own biases (recall issues, social desirability, and lack of momentary awareness). While physiological signals offer objectivity, their validity as direct, unambiguous markers of psychological stress is still debated, as they reflect broader ANS activity.

6.2. Generalizability, Scalability, and Real-World Deployment Hurdles

- **Generalizability:** Models developed using data from one group of individuals often fail to perform well on others due to inter-individual differences in physiology and stress responses. Many studies utilize small, homogenous samples (e.g., university students), limiting the applicability of findings to broader, more diverse populations (different age, ethnicity, and health status). The lack of large-scale, diverse, publicly available datasets collected under varied real-world conditions is a significant bottleneck hindering the development of truly generalizable models [13].
- **Scalability:** Deploying sophisticated AI models, especially deep learning algorithms, for potentially millions of users presents significant technical and logistical challenges. These include managing the vast number of data generated, ensuring sufficient computational resources (cloud infrastructure costs and on-device processing limitations), maintaining system performance and reliability at scale, and providing adequate user support [4].
- **Deployment Hurdles:** There is often a gap between research prototypes and robust, user-friendly products ready for real-world deployment. Practical issues such as the limited battery life of wearable devices, unreliable wireless connectivity, device durability, the need for frequent software updates, and seamless integration into users’ existing digital ecosystems and daily routines must be addressed for successful adoption [4].

6.3. Data Security, Privacy, and Ethical Imperatives

The data collected by stress monitoring systems—continuous physiological signals, behavioral patterns, location, and potentially even facial expressions or voice recordings—are exceptionally sensitive and personal. This raises critical concerns regarding security, privacy, and ethics:

- **Security Risks:** IoT devices are notoriously vulnerable to security breaches. Potential threats include unauthorized access to sensitive health data, data theft, manipulation of sensor readings (which could lead to incorrect diagnoses or interventions), denial-of-service attacks, and device hijacking. Robust security measures are essential, including strong data encryption both at rest and in transit (e.g., using established algorithms like

AES, RSA, or attribute-based encryption (ABE) [39]), secure authentication protocols for users and devices, granular access control mechanisms [40], and secure software development practices. Frameworks specifically designed for the healthcare IoT, such as security context frameworks [41] or privacy-preserving architectures like PP-NDNOT for Named Data Networking [40], aim to address these challenges.

- **Privacy Concerns:** Users may feel uncomfortable with the idea of the continuous, pervasive monitoring of their physiological and behavioral states. Concerns exist about how these data will be used, stored, and potentially shared, for instance, with employers or insurance companies. Transparency regarding data handling practices, clear and informed consent processes providing users with meaningful control over their data, and strict adherence to data protection regulations (e.g., GDPR in Europe and HIPAA in the US) are paramount to building user trust [4].
- **Ethical Considerations:** Beyond security and privacy, several ethical issues arise. Algorithmic bias, stemming from unrepresentative training data, could lead to systems performing poorly for certain demographic groups, potentially exacerbating health disparities. Questions of accountability arise if the system provides incorrect feedback or harmful interventions [32]. The potential impact on user autonomy (e.g., feeling overly reliant on the system) and the risk of inducing anxiety through constant monitoring (“quantified self” stress) must be carefully considered [8].

Implementing the necessary robust security and privacy measures often involves trade-offs with usability. Complex encryption key management, multi-factor authentication procedures, or frequent permission requests, while enhancing security [39], can introduce friction into the user experience, making the system cumbersome or difficult to use. This friction can deter user adoption and adherence, ultimately undermining the system’s intended benefits. Developers face the critical design challenge of finding an optimal balance: implementing strong security and privacy protections in a way that is as transparent and seamless as possible for the end-user, perhaps through context-aware security policies or user-friendly control interfaces, thereby minimizing the negative impact on usability [4].

Table 4. Summary of major challenges in IoT/AI stress monitoring and potential mitigation strategies.

Challenge Category	Specific Challenge Description	Potential Mitigation Strategies/Future Research Directions
Accuracy/Reliability	Lower performance in real-world vs. lab settings; motion artifacts; confounding factors (activity)	Robust artifact detection/removal algorithms; advanced signal processing; multimodal fusion incorporating context (activity and environment); more ecologically valid datasets (e.g., SWEET) [12–14]
	Difficulty establishing reliable ground truth for stress	Combining objective physiological data with EMA/subjective reports; Unsupervised/semi-supervised learning to reduce reliance on labels; standardized labeling protocols [12]
Generalizability/Scalability	Models fail to generalize across individuals/populations/contexts	Larger, diverse public datasets; Federated learning (train on decentralized data); transfer learning; adaptive/personalized models; standardized evaluation protocols [13,19]
	Computational/resource constraints for large-scale deployment	Efficient algorithms (model compression and quantization); edge/fog computing architectures; optimized cloud infrastructure; scalable data management platforms [4]

Table 4. Cont.

Challenge Category	Specific Challenge Description	Potential Mitigation Strategies/Future Research Directions
Privacy/Security/Ethics	Vulnerability of IoT devices and sensitive health data	Strong encryption (end-to-end); secure authentication/access control; regular security audits; privacy-preserving computation (e.g., differential privacy and homomorphic encryption) [4,39–41]
	User privacy concerns regarding continuous monitoring and data usage	Transparent data policies; granular user controls/consent; data minimization; on-device processing where feasible; adherence to regulations (GDPR and HIPAA) [4]
	Algorithmic bias; accountability; impact on autonomy; potential for induced anxiety	Bias detection/mitigation techniques; Explainable AI (XAI); clear ethical guidelines; user-centered design focusing on empowerment; studies on long-term psychological impact [8,13,19,32]
Integration/Usability	Difficulty integrating diverse hardware/software components	Standardized APIs and data formats; modular system design; open-source platforms/frameworks [4]
	Lack of interoperability between devices/platforms; device fragmentation	Industry standards development; middleware solutions; focus on common communication protocols (e.g., Bluetooth LE) [4]
	Poor user experience (comfort, interface, and feedback); low adherence/adoption	User-centered design (UCD); co-design methodologies; iterative usability testing; clear, actionable feedback design; comfortable/aesthetic wearable design [9,19]
Personalization	Balancing personalized accuracy with the need for generalizable models	Hybrid models (general pre-training + personal fine-tuning); adaptive algorithms; federated learning with personalization layers; multi-task learning [13,19,31]
	Acquiring sufficient personalized data for robust models	Self-supervised learning (SSL) on unlabeled data; data augmentation techniques; efficient active learning strategies to request labels strategically [19]

6.3.1. Privacy-Preserving Machine Learning: Federated Learning and Differential Privacy

Beyond basic security [42–44], privacy-preserving machine learning (PPML) techniques are essential to the ethical analysis of health data.

Federated Learning (FL) [45,46] is a decentralized training paradigm where, instead of centralizing sensitive patient data, the AI model is sent to local institutions (e.g., hospitals) to be trained on their local data. Only aggregated, anonymized model updates are sent back to a central server to create an improved global model. This approach allows multiple institutions to collaborate to train more robust and generalizable models on larger, more diverse datasets without directly sharing raw patient data, thereby preserving institutional and patient privacy, as described in Figure 3.

Differential Privacy (DP) [47] offers a mathematical, quantifiable guarantee of individual privacy. It works by adding a carefully calibrated amount of statistical noise to the data, queries, or model outputs. This makes it computationally infeasible to determine whether any particular individual's data were used in the training dataset, thus protecting against re-identification attacks. However, DP introduces a fundamental trade-off: stronger privacy (more noise) often comes at the cost of lower model utility or accuracy.

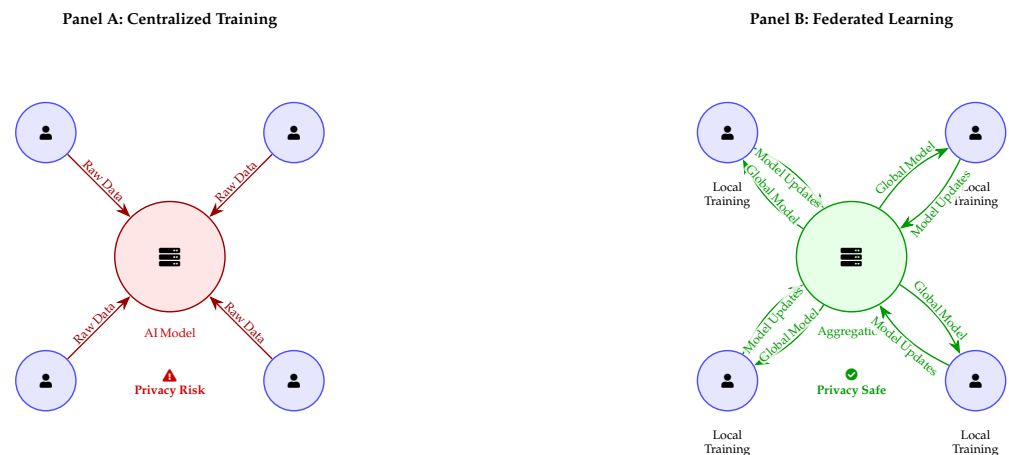


Figure 3. Comparison between centralized training (Panel (A)) and federated learning (Panel (B)), highlighting privacy implications.

6.3.2. The Right to Be Forgotten: Machine Unlearning and Model Editing in Healthcare AI

Two cutting-edge techniques are emerging to address the need to modify models post-training, a crucial capability for AI governance:

- Machine Unlearning directly addresses the GDPR’s “right to be forgotten.” It is the process of selectively removing the influence of a specific data point or user from an already trained model, without requiring a costly and time-consuming retraining from scratch [48,49]. This is vital in healthcare, where a patient may withdraw consent for their data to be used. Research in this area is advancing rapidly, with methods like “Gradient Transformation” being proposed for efficient unlearning on dynamic graphs, a problem analogous to time-series sensor data.
- Model Editing refers to the ability to directly modify a trained model’s behavior to correct erroneous predictions, remove biases, or align the model with expert knowledge. In healthcare, this is critical. For instance, if a sepsis risk model learns a spurious correlation (e.g., that asthma reduces the risk of mortality), clinicians must be able to intervene and correct this behavior without invalidating the rest of the model’s knowledge. Interactive tools like GAM Changer allow domain experts (clinicians) to visualize and “inject” their knowledge directly into the model, enhancing safety and trust [50,51].

The integration of these techniques reveals a complex interdependence. The pursuit of privacy is not a simple fix; it introduces a fundamental tension among three often-competing goals: *privacy*, *utility (accuracy)*, and *fairness*. For example, the strict application of differential privacy has been shown to reduce model accuracy disproportionately for underrepresented demographic groups in the dataset, thereby exacerbating algorithmic biases. Similarly, federated learning can struggle with heterogeneous data from different institutions, affecting the convergence and fairness of the global model. Therefore, the development of responsible AI systems for mental health is not about maximizing a single objective but about carefully navigating this trilemma—a significant challenge that requires further investigation.

6.4. Technological Integration, Interoperability, and Usability

Creating a functional stress monitoring system requires the seamless integration of diverse technological components: multiple sensors, wearable hardware, mobile applications, wireless communication protocols (Bluetooth and Wi-Fi), cloud platforms for data storage and analysis, and AI algorithms. Ensuring these components work together reliably and efficiently is a significant engineering challenge.

The lack of standardization poses a major barrier to interoperability. Different wearable devices may use proprietary data formats or communication protocols, making it difficult to integrate data from multiple vendors or switch between devices. This “device fragmentation” complicates development and limits user choice. Establishing common standards for data representation and communication Application Programming Interfaces (APIs) would greatly facilitate the development of more flexible and integrated systems.

Ultimately, user experience (UX) is paramount to the success of these systems. Devices must be comfortable to wear continuously, interfaces must be intuitive and easy to navigate, setup should be straightforward, and the feedback provided must be clear, understandable, and perceived as valuable by the user. Poor usability, confusing feedback, or a high perceived burden (e.g., frequent charging and complex interactions) will likely lead to low user adherence and eventual abandonment of the technology, regardless of its technical sophistication [19]. Context-aware design, where the system adapts its behavior based on the user’s current situation (e.g., simplifying notifications during activity), can enhance usability [9].

6.5. Balancing Personalization and Generalization

The field faces an ongoing tension between the need for personalization and the desire for generalizable models. Highly personalized models, trained on individual-specific data, can potentially achieve high accuracy for that person but may require significant individual data collection and might not generalize well even to slightly different contexts for the same individual [19]. Conversely, general models trained on large, diverse datasets aim for broader applicability but may fail to capture the crucial nuances of an individual’s unique stress response profile, leading to suboptimal performance at the individual level. Finding the right balance is key. Hybrid approaches, such as models that start with general knowledge but adapt and personalize over time as more individual data become available (e.g., through transfer learning or adaptive algorithms), represent a promising direction. Multi-task learning, which allows models to learn shared representations while also specializing for individual tasks (or users), is another potential avenue [31].

6.6. User Experience and Acceptance Factors

The long-term adoption of stress monitoring technology depends on more than just accuracy and basic usability. Deeper user experience and acceptance factors come into play [4]:

- **Perceived Usefulness:** Users must believe that the system provides tangible benefits, such as increased self-awareness, effective stress reduction, or improved well-being.
- **Trust:** Users need to trust the system’s accuracy, reliability, and critically, its handling of their sensitive data. Transparency in how the system works and how data are used are essential to building trust.
- **Intrusiveness:** Both the monitoring process (wearing sensors) and the interventions delivered must not be overly intrusive or disruptive to daily life.
- **Feedback Quality:** The information provided back to the user must be meaningful, actionable, and easy to understand. Simply presenting raw physiological data is unlikely to be helpful for most users.
- **Aesthetics and Comfort:** The physical design of wearable devices plays a significant role in acceptance.
- **Resistance to Tracking:** Some users may inherently resist the idea of continuous biometric monitoring due to privacy concerns or a feeling of being constantly evaluated [2].

Addressing these factors requires a user-centered design philosophy, involving potential users throughout the development process (co-design) to ensure that the final system meets their needs, preferences, and concerns.

6.7. Analysis of Failure Cases and Contradictory Findings

To address the critique that the review lacks a balanced perspective, it is essential to examine failures and contradictions in the literature. There is a stark disparity between lab and real-world results [52]. While some studies report classification accuracies exceeding 95% or even 99% on specific, controlled datasets [18], studies conducted “in the wild” often yield much more modest results. For example, one study using real-life data from the SWEET dataset achieved an F1-score of only 0.43 [53], and another using PPG and contextual data on university students reached an F1-score of 70% [54].

This gap is driven by multiple factors. Models often fail to generalize across different contexts. One study that evaluated EEG models trained on one stress task (mental arithmetic) and tested on another (VR gaming) found a significant drop in performance, demonstrating that models learn stressor-specific patterns that do not transfer well [55]. Similarly, cross-device generalization is an issue, as seen in studies where models perform differently on devices like the Polar H10 and Empatica E4, even under controlled lab conditions [56]. The literature’s emphasis on promising directions often overlooks these generalization failures, leading to overly optimistic conclusions. A more nuanced assessment of the field requires a transparent acknowledgment of these negative and contradictory findings.

7. Future Research Directions

Although recent advances in the field indicate a sufficient level of maturity to generate a greater societal impact, several barriers continue to hinder its widespread implementation. To address these challenges and unlock the full potential of the IoT and AI in stress management, this section outlines several key research directions that should be prioritized in future work.

7.1. Enhancing Ecological Validity and Robustness Under Free-Living Conditions

There is a critical need to shift focus from controlled laboratory studies towards research conducted in naturalistic, “in-the-wild” settings. This requires the development and utilization of more datasets collected during daily life, capturing the variability and complexity of real-world stressors and contexts (following examples like the SWEET dataset [14]). Concurrently, efforts must concentrate on developing signal processing and machine learning algorithms that are inherently more robust to the noise, motion artifacts, and confounding factors prevalent in free-living data [12]. This includes improving artifact detection and correction techniques and designing models that can explicitly account for or filter out the influence of physical activity and other non-stress-related physiological variations. Furthermore, deeper investigation and the automatic integration of contextual information—such as location, social setting, ongoing activity, time of day, and even environmental factors—are necessary to interpret physiological signals accurately and understand the true drivers of stress responses [12].

7.2. Advancing Model Interpretability (Explainable AI—XAI)

The “black box” nature of many sophisticated AI models, particularly deep learning architectures, remains a significant barrier to clinical acceptance and user trust. Future research must prioritize the development and application of Explainable AI (XAI) techniques tailored for physiological time-series data. Methods like SHapley Additive exPlanations (SHAP), Local Interpretable Model-agnostic Explanations (LIME), attention

mechanisms within models, and rule-based extraction can help elucidate which specific features, signal patterns, or contextual factors are driving a model's stress prediction. Providing interpretable outputs is crucial not only to debugging and improving models but also to enabling clinicians to understand and validate the system's reasoning and empowering users with more transparent insights into their own stress patterns [19].

7.3. Co-Designing Ethical, User-Centered Systems

Moving forward, a paradigm shift towards truly user-centered and ethical design practices is essential. This involves adopting participatory design or co-design methodologies that actively involve all relevant stakeholders—including end-users (from diverse demographic backgrounds), clinicians, caregivers, and ethicists—throughout the entire system development lifecycle, from conception to deployment and evaluation. This ensures that the technology aligns with real user needs, preferences, values, and contexts of use. Concurrently, the research community must work towards establishing clearer ethical guidelines and best practices specifically for the development and deployment of mental health monitoring technologies, addressing data governance, informed consent, algorithmic transparency, bias mitigation, and accountability. The focus should be on creating systems that empower individuals with self-knowledge and tools for well-being, rather than systems that merely monitor or control [9].

7.4. Deepening the Integration of Causal Inference and Contextual Understanding

To overcome the problem of confounders (e.g., physical activity), research must advance from correlational models toward causal inference models. This involves not only integrating rich, multimodal contextual data (e.g., location, activity, social setting, and calendar) but also developing models that can reason about the causal relationships between context, physiology, and stress. The goal is to build systems that, rather than merely correlating patterns, can answer the question, “Is this physiological response *caused* by psychological stress or by physical activity?”

7.5. Exploring Novel Sensing and Algorithmic Frontiers

Continued innovation in both sensing hardware and analytical algorithms is vital. Research should explore the potential of novel or less common sensor modalities. While currently invasive, advancements in non-invasive biochemical sensing (e.g., cortisol in sweat [1]) could provide more direct stress markers if usability improves. Advanced thermal imaging techniques [5], ear-EEG for more discreet brain monitoring, or novel behavioral sensors integrated into everyday objects warrant further investigation. On the algorithmic front, exploring cutting-edge AI techniques beyond standard supervised learning holds promise. Self-supervised learning can leverage a vast number of unlabeled physiological data for representation learning [19]. Federated learning offers a privacy-preserving approach to train models on decentralized data residing on users' devices. Reinforcement learning is particularly relevant for optimizing adaptive intervention strategies in real time [20]. Graph Neural Networks could be employed to model complex relationships between physiological signals, context, and social interactions.

7.6. Need for Longitudinal Studies and Clinical Integration

To truly understand stress dynamics and the effectiveness of interventions, short-term studies are insufficient. There is a pressing need for longitudinal research that tracks individuals over extended periods (weeks, months, or even years) [19]. This will allow for the investigation of stress adaptation, recovery patterns, the long-term impact of chronic stress, and sustained efficacy and engagement with technology-based interventions. Crucially, the gap between research prototypes and clinically validated tools must be

bridged. This requires conducting rigorous clinical trials (e.g., RCTs) to evaluate the real-world effectiveness of these systems in improving mental health outcomes, reducing stress-related symptoms, and promoting positive health behaviors [19]. Furthermore, research should investigate practical strategies for integrating these technologies into existing clinical workflows and healthcare systems to ensure they complement, rather than disrupt, traditional care pathways [4].

7.7. Benchmarking Advanced Architectures for In-the-Wild Generalization

There is a critical need to move beyond proof-of-concept studies and establish rigorous benchmarking protocols. Future research should focus on systematically evaluating advanced model architectures, such as Transformers versus LSTM and hybrid models, on large-scale, longitudinal, “in-the-wild” datasets like SWEET and TILES [57,58]. The goal should not be merely to maximize accuracy but to comprehensively assess the trade-offs between predictive performance, computational cost, inference latency, and suitability for on-device (edge) versus cloud-based deployment. This will provide the community with practical guidelines for selecting the right architecture for specific real-world applications.

7.8. Establishing Robust Evaluation Protocols and Metrics for Real-World Studies

The field must converge on standardized evaluation protocols for “in-the-wild” studies. This includes adopting cross-validation methods that respect temporal order and subject independence, such as leave-one-subject-out (LOSO) cross-validation. Furthermore, given the inherent imbalance of real-life stress data (non-stress moments typically far outnumber stress moments), the community must move beyond simple accuracy. More robust and informative metrics for imbalanced data, such as *Balanced Accuracy (BACC)* and the *Matthews Correlation Coefficient (MCC)*, which are already being used in cutting-edge in-the-wild studies, should be prioritized [59]. Adopting these metrics will allow for a more meaningful and realistic comparison of model performance across studies.

8. Conclusions

8.1. Synthesis of Findings: Progress and Pitfalls

The integration of the IoT and AI has undeniably catalyzed significant advancements in the field of stress monitoring and management. Objective, continuous data collection with wearable sensors, coupled with the analytical power of machine learning and deep learning, enables the detection of physiological and behavioral correlates of stress with increasing accuracy, at least under controlled conditions. Multimodal fusion techniques are enhancing robustness, while personalization strategies are beginning to address the critical issue of inter-individual variability. Furthermore, the development of real-time intervention frameworks, such as JITAI and neurofeedback systems, signals a move towards proactive and dynamic stress modulation. However, despite this progress, substantial pitfalls remain. The translation of these technologies from the laboratory to the complexities of real-world, free-living environments continues to be hampered by challenges related to data quality, confounding factors, and model robustness. Generalizability across diverse populations and contexts remains limited, largely due to the scarcity of large-scale, ecologically valid datasets. Paramount concerns surrounding data security, user privacy, and ethical implications (including algorithmic bias and transparency) must be rigorously addressed to foster user trust and ensure responsible innovation. Finally, issues of technological integration, interoperability, and crucially, user experience and long-term acceptance represent significant hurdles to widespread adoption and demonstrable clinical impact.

8.2. The Transformative Potential of the IoT and AI in Stress Management

Notwithstanding the challenges, the transformative potential of the IoT and AI in mental healthcare, particularly for stress management, is immense. These technologies offer the prospect of shifting the paradigm from infrequent, subjective assessments and reactive treatments towards continuous, objective monitoring and personalized, proactive interventions. Future systems, if developed responsibly, could seamlessly integrate into individuals' daily lives, providing timely self-awareness, tailored coping strategies, and adaptive support precisely when needed. By empowering individuals with deeper insights into their own stress patterns and effective tools for regulation, these technologies could significantly contribute to preventing the negative health consequences of chronic stress and promoting overall mental well-being and resilience.

8.3. Concluding Remarks on Responsible Innovation

Realizing this transformative potential requires a concerted and interdisciplinary effort. Technological innovation in sensors, algorithms, and system architectures must proceed hand in hand with rigorous validation in real-world settings. Crucially, this advancement must be guided by a strong commitment to ethical principles, prioritizing user privacy, data security, algorithmic fairness, and transparency. A user-centered approach involving co-design and the continuous evaluation of usability and acceptance is non-negotiable. Ultimately, the success of the IoT and AI in stress management will be measured not only by technical sophistication but by the ability to deliver robust, ethical, and user-centric solutions that demonstrably improve people's lives. Continued collaboration among engineers, computer scientists, psychologists, clinicians, ethicists, and end-users is essential to navigating the complexities and ensuring that innovation serves genuine human needs responsibly.

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Abbreviations

The following abbreviations are used in this manuscript:

AI	Artificial Intelligence
ANS	Autonomic Nervous System
ACC	accelerometer
AOA	Archimedes Optimization Algorithm
AHP	Analytical Hierarchical Process
ABE	attribute-based encryption
AUC	Area Under the ROC Curve
AU	Action Unit
BVP	Blood Volume Pulse
Bi-LSTM	Bidirectional Long Short-Term Memory
CEP	Complex Event Processing
CNN	Convolutional Neural Network

DL	deep learning
DT	decision tree
ECG	electrocardiogram
EDA	electrodermal activity
EEG	electroencephalogram
EHR	electronic health record
EMA	ecological momentary assessment
EMG	electromyogram
FIS	Fuzzy Inference System
FSR	Force Sensitive Resistor
GBM	Gradient Boosting Machine
GSR	Galvanic Skin Response
HCI	human–computer interaction
HF	high frequency (HRV band)
HRV	heart rate variability
IoT	Internet of Things
ICA	Independent Component Analysis
JITAI	just-in-time adaptive intervention
k-NN	k-nearest neighbors
LBP	Local Binary Patterns
LF	low-frequency (HRV band)
LIMEs	Local Interpretable Model-agnostic Explanations
LOSO	leave one subject out
LR	Logistic Regression
LSTM	Long Short-Term Memory
ML	machine learning
mRMR	Minimum Redundancy Maximum Relevance
NB	Naive Bayes
PANAS	Positive and Negative Affect Schedule
PCA	Principal Component Analysis
PPG	photoplethysmography
PSD	Power Spectral Density
RCT	randomized controlled trial
RF	Random Forest
RL	reinforcement learning
RMSSD	Root Mean Square of Successive Differences
RNN	Recurrent Neural Network
ROC	Receiver Operating Characteristic
RSP	respiration
SCR	skin conductance response
SD	standard deviation
SDNN	standard deviation of NN intervals
SHAP	SHapley Additive exPlanations
SKT	skin temperature
SMOTE	Synthetic Minority Over-sampling Technique
SOM	Self-Organizing Map
SSL	self-supervised learning
STAI	State-Trait Anxiety Inventory
SVM	Support Vector Machine
UX	user experience
WPT	Wavelet Packet Transform
XAI	Explainable Artificial Intelligence
XGBoost	Extreme Gradient Boosting

References

- Lu, P.; Zhang, W.; Ma, L.; Zhao, Q. A Framework of Real-Time Stress Monitoring and Intervention System. In *Proceedings of the Cross-Cultural Design. Applications in Health, Learning, Communication, and Creativity*; Rau, P.L., Ed.; Springer International Publishing: Cham, Switzerland, 2020; pp. 166–175.
- IoT in Wearables 2025: Devices, Examples and Industry Overview. Available online: <https://stormotion.io/blog/iot-in-wearables/> (accessed on 3 May 2025).
- Abd Al-Alim, M.; Mubarak, R.; Salem, N.M.; Sadek, I. A Machine-Learning Approach for Stress Detection Using Wearable Sensors in Free-Living Environments. *Comput. Biol. Med.* **2024**, *179*, 108918. [\[CrossRef\]](#) [\[PubMed\]](#)
- Mohamad Jawad, H.; Bin Hassan, Z.; Zaidan, B.; Mohammed Jawad, F.; Mohamed Jawad, D.; Alredany, W. A Systematic Literature Review of Enabling IoT in Healthcare: Motivations, Challenges, and Recommendations. *Electronics* **2022**, *11*, 3223. [\[CrossRef\]](#)
- Mattioli, V.; Davoli, L.; Belli, L.; Gambetta, S.; Carnevali, L.; Sgoifo, A.; Raheli, R.; Ferrari, G. IoT-Based Assessment of a Driver's Stress Level. *Sensors* **2024**, *24*, 5479. [\[CrossRef\]](#) [\[PubMed\]](#)
- Muñoz Arteaga, J.; Hernández, Y. *Temas de Diseño En Interacción Humano-Computadora*; Iniciativa Latinoamericana de Libros de Texto Abiertos (LATIn): São Paulo, Brazil, 2014.
- Biggs, A.; Brough, P.; Drummond, S. Lazarus and Folkman's Psychological Stress and Coping Theory. In *The Handbook of Stress and Health*; John Wiley & Sons, Ltd.: Chichester, UK, 2017; pp. 349–364.
- Pei, G.; Li, H.; Lu, Y.; Wang, Y.; Hua, S.; Li, T. Affective Computing: Recent Advances, Challenges, and Future Trends. *Intell. Comput.* **2024**, *3*, 0076. [\[CrossRef\]](#)
- User Experience: UX in IoT: Designing for a Connected Experience. Available online: <https://fastercapital.com/content/User-experience--UX---UX-in-IoT--UX-in-IoT--Designing-for-a-Connected-Experience.html> (accessed on 3 May 2025).
- Cai, Y. Empathic Computing. In *Ambient Intelligence in Everyday Life: Foreword by Emile Aarts*; Cai, Y., Abascal, J., Eds.; Springer: Berlin/Heidelberg, Germany, 2006; pp. 67–85.
- Roshanaei, M.; Rezapour, R.; El-Nasr, M. Talk, Listen, Connect: Navigating Empathy in Human-AI Interactions. *arXiv* **2024**, arXiv:2409.15550.
- Lazarou, E.; Exarchos, T. Predicting Stress Levels Using Physiological Data: Real-Time Stress Prediction Models Utilizing Wearable Devices. *AIMS Neurosci.* **2024**, *11*, 76–102. [\[CrossRef\]](#)
- Vos, G.; Trinh, K.; Sarnyai, Z.; Azghadi, M. Generalizable Machine Learning for Stress Monitoring from Wearable Devices: A Systematic Literature Review. *Int. J. Med. Inform.* **2023**, *173*, 105026. [\[CrossRef\]](#)
- Mozos, O.; Sandulescu, V.; Andrews, S.; Ellis, D.; Bellotto, N.; Dobrescu, R.; Ferrandez, J. Stress Detection Using Wearable Physiological and Sociometric Sensors. *Int. J. Neural Syst.* **2017**, *27*, 1650041. [\[CrossRef\]](#)
- Schmidt, P.; Reiss, A.; Duerichen, R.; Marberger, C.; Van Laerhoven, K. Introducing WESAD, a Multimodal Dataset for Wearable Stress and Affect Detection. In *Proceedings of the 20th ACM International Conference on Multimodal Interaction, ICMI '18*, New York, NY, USA, 16–20 October 2018; pp. 400–408. [\[CrossRef\]](#)
- Luna-Perejón, F.; Montes-Sánchez, J.; Durán-López, L.; Vazquez-Baeza, A.; Beasley-Bohórquez, I.; Sevillano-Ramos, J. IoT Device for Sitting Posture Classification Using Artificial Neural Networks. *Electronics* **2021**, *10*, 1825. [\[CrossRef\]](#)
- Androutsou, T.; Angelopoulos, S.; Hristoforou, E.; Matsopoulos, G.; Koutsouris, D. A Multisensor System Embedded in a Computer Mouse for Occupational Stress Detection. *Biosensors* **2023**, *13*, 10. [\[CrossRef\]](#)
- Talaat, F.; El-Balka, R. Stress Monitoring Using Wearable Sensors: IoT Techniques in Medical Field. *Neural Comput. Appl.* **2023**, *35*, 18571–18584. [\[CrossRef\]](#) [\[PubMed\]](#)
- Bolpagni, M.; Pardini, S.; Dianti, M.; Gabrielli, S. Personalized Stress Detection Using Biosignals from Wearables: A Scoping Review. *Sensors* **2024**, *24*, 3221. [\[CrossRef\]](#) [\[PubMed\]](#)
- Joseph, J.; Judy, M. Dynamic Emotion Regulation through Reinforcement Learning: A Neurofeedback System with EEG Data and Custom Wearable Interventions. In *Proceedings of the 2024 International Conference on Brain Computer Interface & Healthcare Technologies (iCon-BCIHT)*, IEEE, Thiruvananthapuram, India, 19–20 December 2024; pp. 169–176.
- Patil, S.; Paithane, A. Optimized EEG-Based Stress Detection: A Novel Approach. *Biomed. Pharmacol. J.* **2024**, *17*, 2607–2616. [\[CrossRef\]](#)
- Gkintoni, E.; Aroutzidis, A.; Antonopoulou, H.; Halkiopoulos, C. From Neural Networks to Emotional Networks: A Systematic Review of EEG-Based Emotion Recognition in Cognitive Neuroscience and Real-World Applications. *Brain Sci.* **2025**, *15*, 220. [\[CrossRef\]](#)
- Ballesteros, J.A.; Ramírez V, G.M.; Moreira, F.; Solano, A.; Pelaez, C.A. Facial Emotion Recognition through Artificial Intelligence. *Front. Comput. Sci.* **2024**, *6*, 1359471. [\[CrossRef\]](#)
- Ma, F.; Sun, B.; Li, S. Facial Expression Recognition with Visual Transformers and Attentional Selective Fusion. *IEEE Trans. Affect. Comput.* **2023**, *14*, 1236–1248. [\[CrossRef\]](#)

25. Hindu, A.; Bhowmik, B. An IoT-Enabled Stress Detection Scheme Using Facial Expression. In Proceedings of the 2022 IEEE 19th India Council International Conference (INDICON), Kerala, India, 24–26 November 2022; pp. 1–6.
26. WESAD (Wearable Stress and Affect Detection) Dataset. Available online: <https://www.kaggle.com/datasets/orvile/wesad-wearable-stress-affect-detection-dataset> (accessed on 3 May 2025).
27. AMIGOS: A Dataset for Affect, Personality and Mood Research on Individuals and Groups. Available online: <http://www.eecs.qmul.ac.uk/mmv/datasets/amigos/> (accessed on 3 May 2025).
28. Li, H.; Sun, G.; Li, Y.; Yang, R. Wearable Wireless Physiological Monitoring System Based on Multi-Sensor. *Electronics* **2021**, *10*, 986. [\[CrossRef\]](#)
29. Nancy, A.; Ravindran, D.; Raj Vincent, P.; Srinivasan, K.; Gutierrez Reina, D. IoT-Cloud-Based Smart Healthcare Monitoring System for Heart Disease Prediction via Deep Learning. *Electronics* **2022**, *11*, 2292. [\[CrossRef\]](#)
30. Al-Atawi, A.; Alyahyan, S.; Alatawi, M.; Sadad, T.; Manzoor, T.; Farooq-i-Azam, M.; Khan, Z. Stress Monitoring Using Machine Learning, IoT and Wearable Sensors. *Sensors* **2023**, *23*, 8875. [\[CrossRef\]](#)
31. Razavi, M.; Ziyadidegan, S.; Jahromi, R.; Kazeminasab, S.; Janfaza, V.; Mahmoudzadeh, A.; Baharlouei, E.; Sasangohar, F. Machine Learning, Deep Learning and Data Preprocessing Techniques for Detection, Prediction, and Monitoring of Stress and Stress-related Mental Disorders: A Scoping Review. *arXiv* **2024**, arXiv:2308.04616. [\[CrossRef\]](#)
32. Moshawrab, M.; Adda, M.; Bouzouane, A.; Ibrahim, H.; Raad, A. Reviewing Multimodal Machine Learning and Its Use in Cardiovascular Diseases Detection. *Electronics* **2023**, *12*, 1558. [\[CrossRef\]](#)
33. Marković, D.; Vujičić, D.; Stojić, D.; Jovanović, Ž.; Pešović, U.; Randić, S. Monitoring System Based on IoT Sensor Data with Complex Event Processing and Artificial Neural Networks for Patients Stress Detection. In Proceedings of the 2019 18th International Symposium INFOTEH-JAHORINA (INFOTEH), Jahorina, Bosnia and Herzegovina, 20–22 March 2019; pp. 1–6.
34. Johnson, J.; Zawadzki, M.; Sliwinski, M.; Almeida, D.; Buxton, O.; Conroy, D.; Marcusson-Clavertz, D.; Kim, J.; Stawski, R.; Scott, S.; et al. Adaptive Just-in-Time Intervention to Reduce Everyday Stress Responses: Protocol for a Randomized Controlled Trial. *JMIR Res. Protoc.* **2025**, *14*, e58985. [\[CrossRef\]](#) [\[PubMed\]](#)
35. Liu, M.; Ren, S.; Ma, S.; Jiao, J.; Chen, Y.; Wang, Z.; Song, W. Gated transformer networks for multivariate time series classification. *arXiv* **2021**, arXiv:2103.14438. [\[CrossRef\]](#)
36. Zhou, H.; Zhang, S.; Peng, J.; Zhang, S.; Li, J.; Xiong, H.; Zhang, W. Informer: Beyond Efficient Transformer for Long Sequence Time-Series Forecasting. In Proceedings of the AAAI Conference on Artificial Intelligence, Vancouver, BC, Canada, 2–9 February 2021; Volume 35, pp. 11106–11115. [\[CrossRef\]](#)
37. Mamieva, D.; Abdusalomov, A.; Kutlimuratov, A.; Muminov, B.; Whangbo, T. Multimodal Emotion Detection via Attention-Based Fusion of Extracted Facial and Speech Features. *Sensors* **2023**, *23*, 5475. [\[CrossRef\]](#)
38. Sun, F.T.; Cynthia, K.; Cheng, H.T.; Buthpitiya, S.; Collins, P.; Griss, M. Activity-Aware Mental Stress Detection Using Physiological Sensors. In *Mobile Computing, Applications, and Services—MobiCASE 2010*; Lecture Notes of the Institute for Computer Sciences, Social Informatics and Telecommunications Engineering; Springer: Berlin/Heidelberg, Germany, 2012; Volume 76. [\[CrossRef\]](#)
39. Irshad, R.; Sohail, S.; Hussain, S.; Madsen, D.; Zamani, A.; Ahmed, A.; Alattab, A.; Badr, M.; Alwayle, I. Towards Enhancing Security of IoT-Enabled Healthcare System. *Heliyon* **2023**, *9*, e22336. [\[CrossRef\]](#)
40. Boussada, R.; Hamdane, B.; Elhdhili, M.; Saidane, L. PP-NDNoT: On Preserving Privacy in IoT-Based E-Health Systems over NDN. In Proceedings of the 2019 IEEE Wireless Communications and Networking Conference (WCNC), Marrakesh, Morocco, 15–18 April 2019; pp. 1–6.
41. The Fusion of Internet of Things, Artificial Intelligence, and Cloud Computing in Health Care. Available online: <https://www.springerprofessional.de/the-fusion-of-internet-of-things-artificial-intelligence-and-clo/19562918> (accessed on 3 May 2025).
42. Al-Garadi, M.A.; Mohamed, A.; Al-Ali, A.K.; Du, X.; Ali, I.; Guizani, M. A Survey of Machine and Deep Learning Methods for Internet of Things (IoT) Security. *IEEE Commun. Surv. Tutor.* **2020**, *22*, 1646–1685. [\[CrossRef\]](#)
43. Weber, R.H. Internet of Things—New security and privacy challenges. *Comput. Law Secur. Rev.* **2010**, *26*, 23–30. [\[CrossRef\]](#)
44. Mittelstadt, B. Designing the Health-Related Internet of Things: Ethical Principles and Guidelines. *Information* **2017**, *8*, 77. [\[CrossRef\]](#)
45. Pati, S.; Kumar, S.; Varma, A.; Edwards, B.; Lu, C.; Qu, L.; Wang, J.J.; Lakshminarayanan, A.; Wang, S.H.; Sheller, M.J.; et al. Privacy preservation for federated learning in health care. *Patterns* **2024**, *5*, 100974. [\[CrossRef\]](#)
46. Li, T.; Sahu, A.K.; Talwalkar, A.; Smith, V. Federated Learning: Challenges, Methods, and Future Directions. *IEEE Signal Process. Mag.* **2020**, *37*, 50–60. [\[CrossRef\]](#)
47. Abadi, M.; Chu, A.; Goodfellow, I.; McMahan, H.B.; Mironov, I.; Talwar, K.; Zhang, L. Deep Learning with Differential Privacy. In Proceedings of the 2016 ACM SIGSAC Conference on Computer and Communications Security, Vienna, Austria, 24–28 October 2016; pp. 308–318. [\[CrossRef\]](#)
48. Zhang, H.; Wu, B.; Yang, X.; Yuan, X.; Liu, X.; Yi, X. Dynamic Graph Unlearning: A General and Efficient Post-Processing Method via Gradient Transformation. In Proceedings of the ACM on Web Conference 2025, Sydney, NSW, Australia, 28 April–2 May 2025; pp. 931–944. [\[CrossRef\]](#)

49. Kumar, V.; Roy, D. Machine Unlearning Models for Medical Care and Health Data Privacy in Healthcare 6.0. In *Exploration of Transformative Technologies in Healthcare 6.0* Eds.; IGI Global Scientific Publishing: Hershey, PA, USA, 2025; pp. 273–302. [\[CrossRef\]](#)
50. Wang, Z.J.; Kale, A.; Nori, H.; Stella, P.; Nunnally, M.; Chau, D.H.; Vorvoreanu, M.; Vaughan, J.W.; Caruana, R. GAM Changer: Editing Generalized Additive Models with Interactive Visualization. *arXiv* **2021**, arXiv:2112.03245. [\[CrossRef\]](#)
51. Meng, K.; Bau, D.; Andonian, A.; Belinkov, Y. Locating and Editing Factual Associations in GPT. *arXiv* **2023**, arXiv:2202.05262. [\[CrossRef\]](#)
52. Mishra, V.; Hao, T.; Sun, S.; Walter, K.N.; Ball, M.J.; Chen, C.H.; Zhu, X. Investigating the Role of Context in Perceived Stress Detection in the Wild. In Proceedings of the 2018 ACM International Joint Conference and 2018 International Symposium on Pervasive and Ubiquitous Computing and Wearable Computers, UbiComp '18, Singapore, 9–11 October 2018; pp. 1708–1716. [\[CrossRef\]](#)
53. Başaran, O.T.; Can, Y.S.; André, E.; Ersoy, C. Relieving the burden of intensive labeling for stress monitoring in the wild by using semi-supervised learning. *Front. Psychol.* **2024**, *14*, 1293513 [\[CrossRef\]](#)
54. Aqajari, S.A.H.; Labbaf, S.; Tran, P.H.; Nguyen, B.; Mehrabadi, M.A.; Levorato, M.; Dutt, N.; Rahmani, A.M. Context-Aware Stress Monitoring using Wearable and Mobile Technologies in Everyday Settings. *arXiv* **2023**, arXiv:2401.05367. [\[CrossRef\]](#)
55. Attallah, O.; Mamdouh, M.; Al-Kabbany, A. Cross-Context Stress Detection: Evaluating Machine Learning Models on Heterogeneous Stress Scenarios Using EEG Signals. *AI* **2025**, *6*, 79. [\[CrossRef\]](#)
56. Amin, O.B.; Mishra, V.; Tapera, T.M.; Volpe, R.; Sathyanarayana, A. Extending Stress Detection Reproducibility to Consumer Wearable Sensors. *arXiv* **2025**, arXiv:2505.05694. [\[CrossRef\]](#)
57. Smets, E.; Rios Velazquez, E.; Schiavone, G.; Chakroun, I.; D'Hondt, E.; De Raedt, W.; Cornelis, J.; Janssens, O.; Van Hoecke, S.; Claes, S.; et al. Large-scale wearable data reveal digital phenotypes for daily-life stress detection. *NPJ Digit. Med.* **2018**, *1*, 67. [\[CrossRef\]](#)
58. Mundnich, K.; Booth, B.M.; L'Hommedieu, M.; Feng, T.; Girault, B.; L'Hommedieu, J.; Wildman, M.; Skaaden, S.; Nadarajan, A.; Villatte, J.L.; et al. TILES-2018, a longitudinal physiologic and behavioral data set of hospital workers. *Sci. Data* **2020**, *7*, 354. [\[CrossRef\]](#)
59. Wang, J.C.; Chien, W.S.; Chen, H.Y.; Lee, C.C. In-The-Wild HRV-Based Stress Detection Using Individual-Aware Metric Learning. In Proceedings of the 2024 IEEE 20th International Conference on Body Sensor Networks (BSN), Chicago, IL, USA, 15–17 October 2024; pp. 1–4. [\[CrossRef\]](#)

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